

Quantitative Risk Management

*Week 1 Introduction to Quantitative Risk
Management and Career Path*

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Lecturer Biography

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5 points of score for personal website and blog on QRM content

Course Outline

Risk Management Process

Risk Management Process: Identification

1. Identification

- ▶ The first step involves identifying potential risks that could affect the organization's objectives.
- ▶ This includes both internal and external risks, such as market fluctuations, operational failures, legal liabilities, and technological changes.
- ▶ Effective risk identification requires a thorough understanding of the business environment and the ability to foresee possible challenges.

Risk Management Process: Assessment

2. Assessment

- ▶ Once risks are identified, they must be assessed in terms of their potential impact and likelihood.
- ▶ This step involves quantitative and qualitative analysis to prioritize risks based on their severity and the organization's risk tolerance.
- ▶ Tools such as risk matrices, scenario analysis, and stress testing are commonly used for risk assessment.

Risk Management Process: Monitoring

3. Monitoring

- ▶ Continuous monitoring of identified risks is essential to detect any changes in their status or impact.
- ▶ This step ensures that risk management strategies remain effective and are adjusted as needed in response to new information or changing circumstances.
- ▶ Regular risk reporting and review meetings are critical components of effective risk monitoring.

Risk Management Process: Control or Mitigation

4. Control or Mitigation

- ▶ The final step involves implementing strategies to control or mitigate the impact of risks.
- ▶ This could include risk avoidance, risk transfer (e.g., through insurance), risk reduction, or acceptance (if the risk is within acceptable limits).
- ▶ Effective risk mitigation requires a balanced approach, ensuring that the costs of risk management are justified by the benefits.

A balanced risk management strategy is key to achieving organizational objectives.

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- ▶ **Liquidity Risk:**

- ▶ The risk that an entity will be unable to meet its short-term financial obligations due to an inability to convert assets into cash without significant loss in value.

Example

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Example: Stock Market Volatility

- ▶ **Scenario:** A financial institution holds a significant portfolio of equities. Sudden market downturns due to economic uncertainty or geopolitical events lead to sharp declines in stock prices.

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- ▶ **Impact:** The institution's portfolio loses value rapidly, potentially resulting in substantial financial losses.
- ▶ **Mitigation:** Diversification of the portfolio across various asset classes and regions, as well as the use of hedging strategies such as options or futures to reduce exposure to market volatility.

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- ▶ **Impact:** The company defaults on its loan, resulting in financial loss for the bank.
- ▶ **Mitigation:** Implementing thorough credit assessments before issuing loans, setting appropriate credit limits, and maintaining adequate collateral requirements. Additionally, the bank can use credit derivatives to hedge against potential losses.

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- ▶ **Mitigation:** Implementing robust cybersecurity measures, including regular system updates, encryption, employee training, and incident response planning. Additionally, the institution can invest in cybersecurity insurance to mitigate financial losses.

Regulation

The Basel Accord: Basel I and Basel II

► **Overview:**

- The Basel Accord refers to a set of international banking regulations developed by the Basel Committee on Banking Supervision (BCBS).
- It provides guidelines for risk management and the minimum capital requirements that financial institutions must hold to mitigate risks.

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► Basel II:

- Implemented in 2004, it expanded on Basel I by incorporating market risk, operational risk, and the three pillars framework:
 1. Minimum capital requirements.
 2. Supervisory review process.
 3. Market discipline.

The Basel Accord: Basel III

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- Aims to improve the banking sector's ability to absorb shocks arising from financial and economic stress.

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▶ **Key Enhancements:**

- ▶ **Capital Requirements:** Higher quality of capital, with an emphasis on common equity.
- ▶ **Liquidity Standards:** Introduction of Liquidity Coverage Ratio (LCR) and Net Stable Funding Ratio (NSFR) to ensure banks maintain adequate liquidity.
- ▶ **Leverage Ratio:** Non-risk-based leverage ratio to serve as a backstop to risk-based measures.

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- Provides a global framework for financial reporting, ensuring consistency, transparency, and comparability of financial statements across countries.

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► Key Objectives:

- To bring transparency by enhancing the international comparability and quality of financial information.
- To strengthen accountability by reducing the information gap between investors and companies.
- To contribute to economic efficiency by helping investors identify opportunities and risks across the world.

Key IFRS Standards

▶ **IFRS 9 - Financial Instruments:**

- ▶ Addresses the classification, measurement, and recognition of financial assets and liabilities.
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▶ **IFRS 16 - Leases:**

- ▶ Introduces a single lessee accounting model, requiring lessees to recognize assets and liabilities for all leases.
- ▶ Eliminates the distinction between operating and finance leases for lessees.

Emerging Risk (Climate): IFRS S2 Climate-related Disclosures

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- ▶ **Scope:** Entities must report on both physical and transition risks related to climate change, as well as any opportunities that may arise.
- ▶ **Key Requirements:**
 - ▶ Disclosure of governance processes related to climate risk management.
 - ▶ Explanation of the entity's strategy for addressing climate-related risks and opportunities.
 - ▶ Detailed processes for identifying, assessing, and managing these risks, and how they integrate with overall risk management.
 - ▶ Performance metrics related to climate-related goals and regulatory requirements.

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 - ▶ Performance metrics related to climate-related goals and regulatory requirements.
- ▶ **Implementation:** Effective from January 2024, with earlier adoption encouraged if IFRS S1 is also applied.

Risk Management Roles

Risk Analyst

- ▶ **Role:** Risk Analysts identify, analyze, and assess the risk associated with financial transactions and investment portfolios.
- ▶ **Responsibilities:**
 - ▶ Evaluating financial risks by analyzing market trends and economic conditions.
 - ▶ Developing risk assessment models to predict potential outcomes.
 - ▶ Preparing risk reports and presenting findings to management.
- ▶ **Skills:** Strong analytical abilities, proficiency in statistical software, and understanding of financial markets.

Risk Manager

- ▶ **Role:** Risk Managers oversee the overall risk management processes within a financial institution or corporation.
- ▶ **Responsibilities:**
 - ▶ Identifying and assessing risks that could impact the organization's objectives.
 - ▶ Developing and implementing risk management strategies and policies.
 - ▶ Ensuring compliance with regulatory requirements and internal policies.
- ▶ **Skills:** Leadership, risk assessment, regulatory knowledge, and strong communication skills.

Actuarial Analyst

- ▶ **Role:** Actuarial Analysts assess the financial risks associated with insurance policies, pensions, and investments.
- ▶ **Responsibilities:**
 - ▶ Using statistical models to analyze the probability and impact of future events.
 - ▶ Calculating insurance premiums and pension contributions based on risk assessments.
 - ▶ Assisting actuaries in developing and pricing new insurance products.
- ▶ **Skills:** Strong mathematical abilities, proficiency in actuarial software, and understanding of insurance and pension systems.

Real Jobs I

- ▶ Market Risk Specialist
<https://www.linkedin.com/jobs/view/3763851034/?>
- ▶ Market Risk Consultant
<https://www.linkedin.com/jobs/view/3994282226/>
- ▶ IRB Risk Modelling Manager
<https://www.linkedin.com/jobs/view/3941425354/>
- ▶ Quantitative Risk Manager
<https://www.linkedin.com/jobs/view/3989346982/>
- ▶ Risk, Model Validation Quant, AVP:
<https://www.linkedin.com/jobs/view/3969796094/>

Real Jobs II

- ▶ Catastrophe Risk Modeller
<https://www.linkedin.com/jobs/view/3975887755/>
- ▶ Climate Change Risk Modeller
<https://www.linkedin.com/jobs/view/3973365600>

Overview of the FRM® Test

Key Details of the FRM® Test

- ▶ **Exam Structure:** The FRM® exam is divided into two parts:
 - ▶ **Part I:** Focuses on the foundational tools of risk management, including quantitative analysis, financial markets, and risk models.
 - ▶ **Part II:** Emphasizes the practical application of risk management tools, covering areas like market risk, credit risk, operational risk, and current industry issues.
- ▶ **Format:** Both parts consist of 100 multiple-choice questions, each with a duration of 4 hours.
- ▶ **Scoring:** Candidates receive a pass/fail grade. Part I must be passed before Part II can be attempted.
- ▶ **Frequency:** The exam is offered twice a year, in May and November, at global testing centers.
- ▶ **Preparation:** The test is rigorous, requiring thorough preparation in all areas covered.

<https://www.garp.org/frm>

Additional Courses for Career

1. Risk Management Specialization: <https://www.coursera.org/specializations/risk-management>
2. Financial Engineering and Risk Management Specialization: <https://www.coursera.org/specializations/financialengineering>